

Micro I, class 19

I began by confirming the standard result that single-output monopolies who charge uniform prices produce less output than an optimal (net-surplus-maximizing) output: specifically if q_0 maximizes monopoly profit and q_1 maximizes net surplus (total dollar benefit minus cost), then $q_1 \geq q_0$. Preferences of the firm ($t = 0$) and planner ($t = 1$) are represented by $f(q, t)$:

$$f(q, t) = \begin{cases} \tilde{p}(q)q - c(q) & \text{for } t = 0 \\ \int_0^q \tilde{p}(y)dy - c(q) & \text{for } t = 1, \end{cases}$$

where $\tilde{p} = d^{-1}$. The conclusion is a simple consequence of Topkis's theorem (and also follows from a revealed-preference argument). I showed that

$$\frac{\partial}{\partial q}[f(q, 1) - f(q, 0)] > 0$$

after confining attention to $\{q \geq 0 \mid \tilde{p}(q) > 0\}$, so f satisfies SID in q and t on the relevant domain. I illustrate the conclusion graphically.

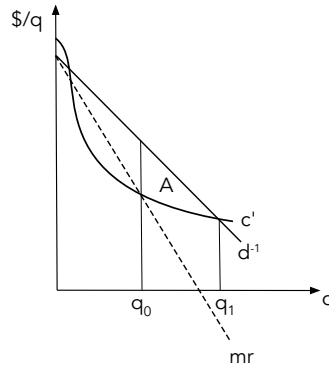


Figure 1: Illustration that a monopolist who charges a uniform price produces less output than is socially optimal. Marginal cost (c') is decreasing, consistent with a monopoly market structure. Neither the firm's nor the planner's problem is concave or quasiconcave (suggesting that using the FOCs to prove the conclusion is apt to fail; in the Figure, both the firm's and planner's problem have *two* interior solutions to their FOCs, but in each case one of them is a strict local minimum). Here marginal revenue is $mr(q) = \tilde{p}(q) + \tilde{p}'(q)q$, profit is $f(q, 0) = \int_0^q f_q(y, 0)dy = \int_0^q [mr(y) - c'(y)]dy > 0$ (the last is true in the figure's example, not generally) and surplus is $f(q, 1) = \int_0^q f_q(y, 1)dy = \int_0^q [d^{-1}(y) - c'(y)]dy > 0$ (again in the example). The *deadweight loss*—the loss of social welfare from monopoly underproduction—is the area $A = \int_{q_0}^{q_1} [d^{-1}(y) - c'(y)]dy$.

I then introduced the notion of a *strategic form game*, and will continue next time. I have published notes on strategic form games on Canvas. I mentioned in class that the Osborne-Rubinstein game theory text is available free from <https://books.osborne.economics.utoronto.ca>. I urge you to download it.